

BRIDGING THE GAP: REAL-TIME QOE MONITORING

Requirement Gathering & Stakeholder Analysis Phase for Mobile Network Optimization

Internet Programming & Mobile Programming

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THE USER GAP

MOBILE NETWORK PROBLEM

USER EXPERIENCE

- Slow internet
- Network instability
- Service interruptions

THE KEY GAP

Existing systems focus on network metrics (QoS), but lack real user experience (QoE).

WE AIM TO BRIDGE THIS GAP.

GATHERING OBJECTIVES



UNDERSTAND PROBLEMS

Pinpoint real user challenges and network failures.



IDENTIFY NEEDS

Determine exactly what stakeholders expect from a solution.



COLLECT DATA

Gather reliable evidence to guide the system design.

KEY STAKEHOLDERS

1. SMARTPHONE USERS

Heavy Usage

Daily dependence on mobile internet for work and life.

Direct Experience

The primary victims of network failures and instability.

2. NETWORK PROVIDERS

MTN, Orange, Camtel

Responsible for network performance and delivery.

Technical View

Utilize mature technical monitoring systems (QoS).

“THEY SEE THE NETWORK DIFFERENTLY.”

GATHERING TECHNIQUES



SURVEY

Google Forms



INTERVIEW

MTN Technology Staff

SURVEY OVERVIEW

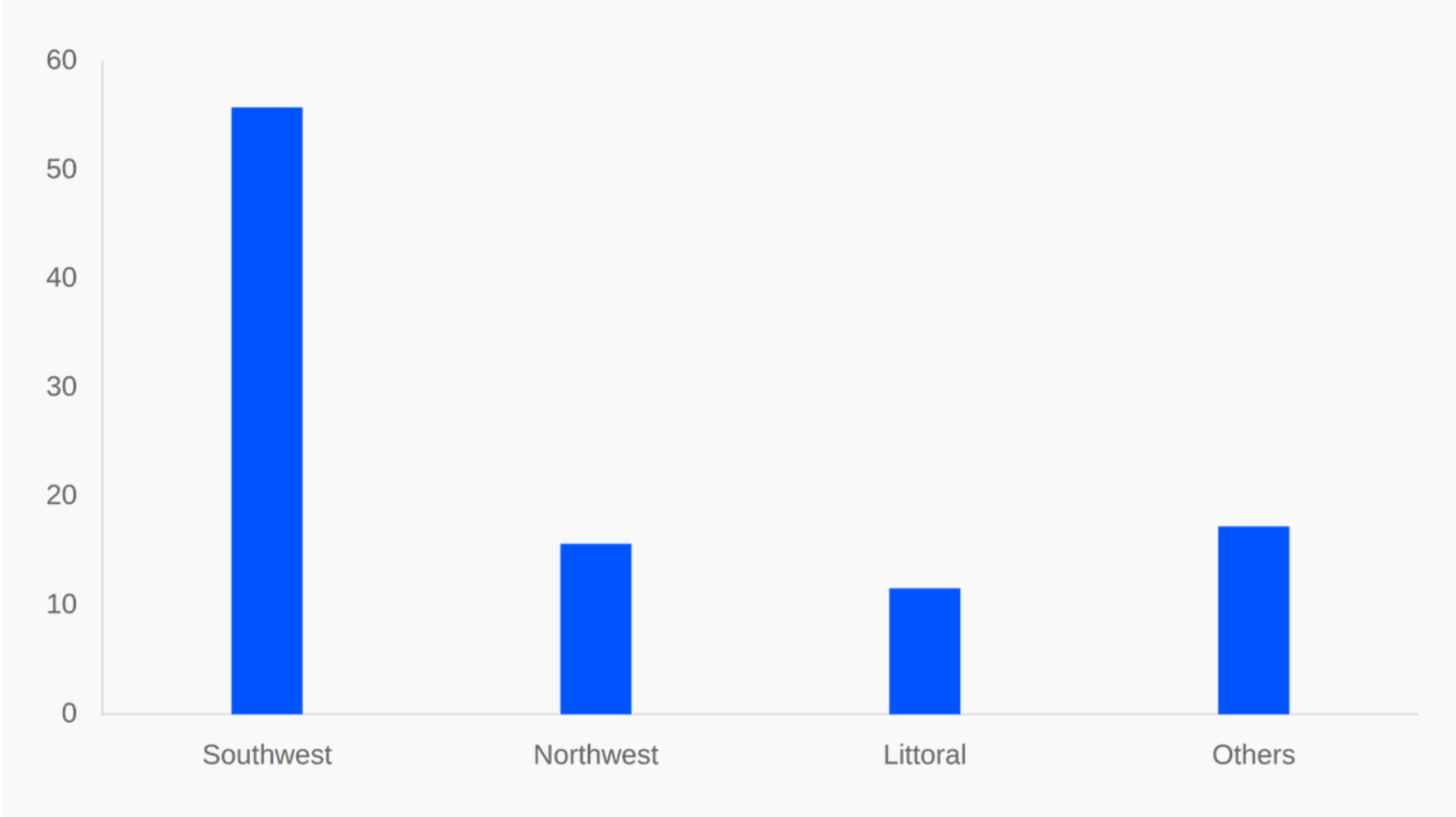
123

VALID RESPONSES

Target: Active Smartphone Users

Focus: Mobile Network Experience & Behavior

Geographical Distribution (%)



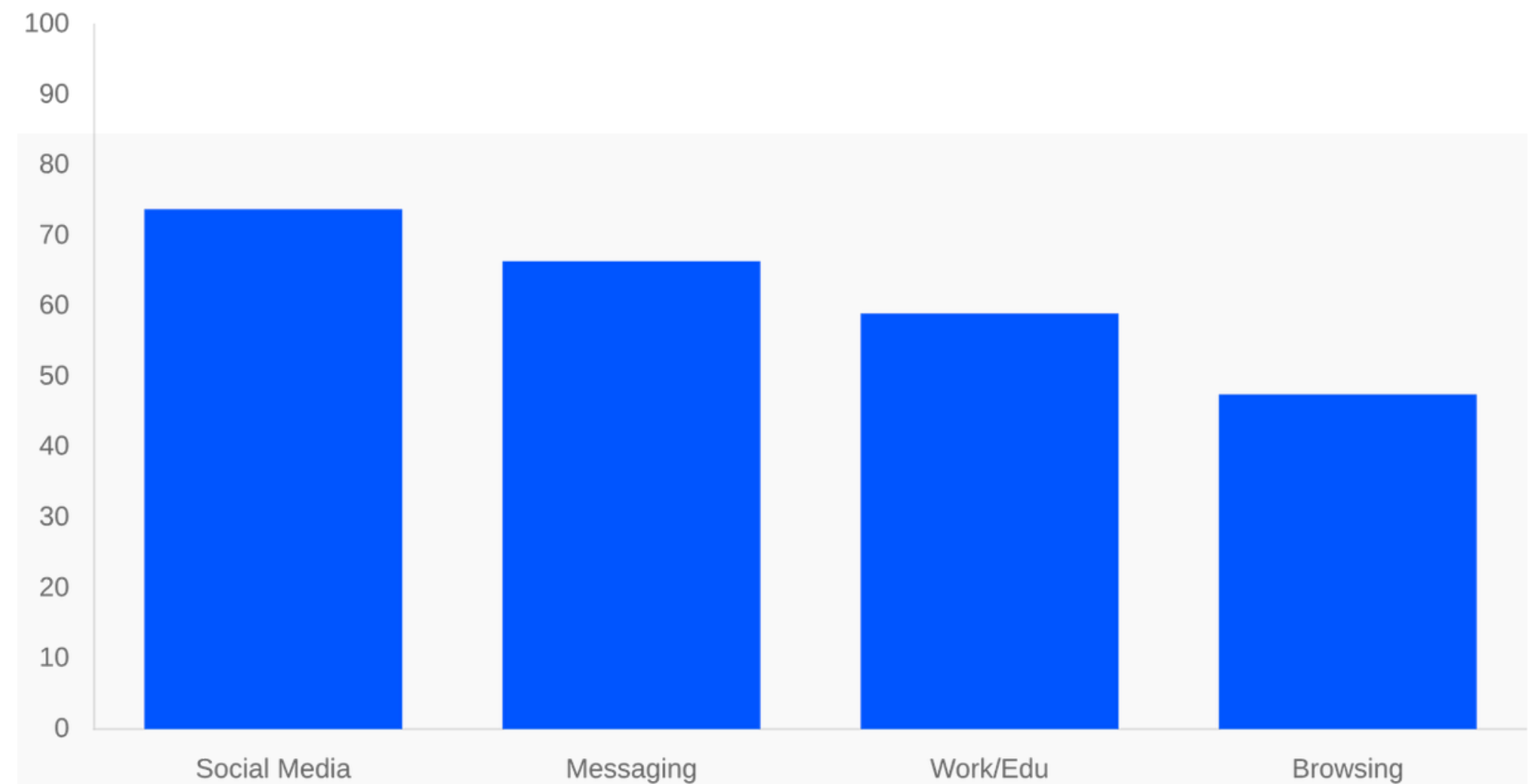
SMARTPHONE USERS

89.4%

High Dependence

Users rely heavily on mobile data for essential activities:

- Social Media (73.8%)
- Messaging (66.4%)
- Work/Education (59.0%)



Data Source: Requirement Gathering Survey (n=122)

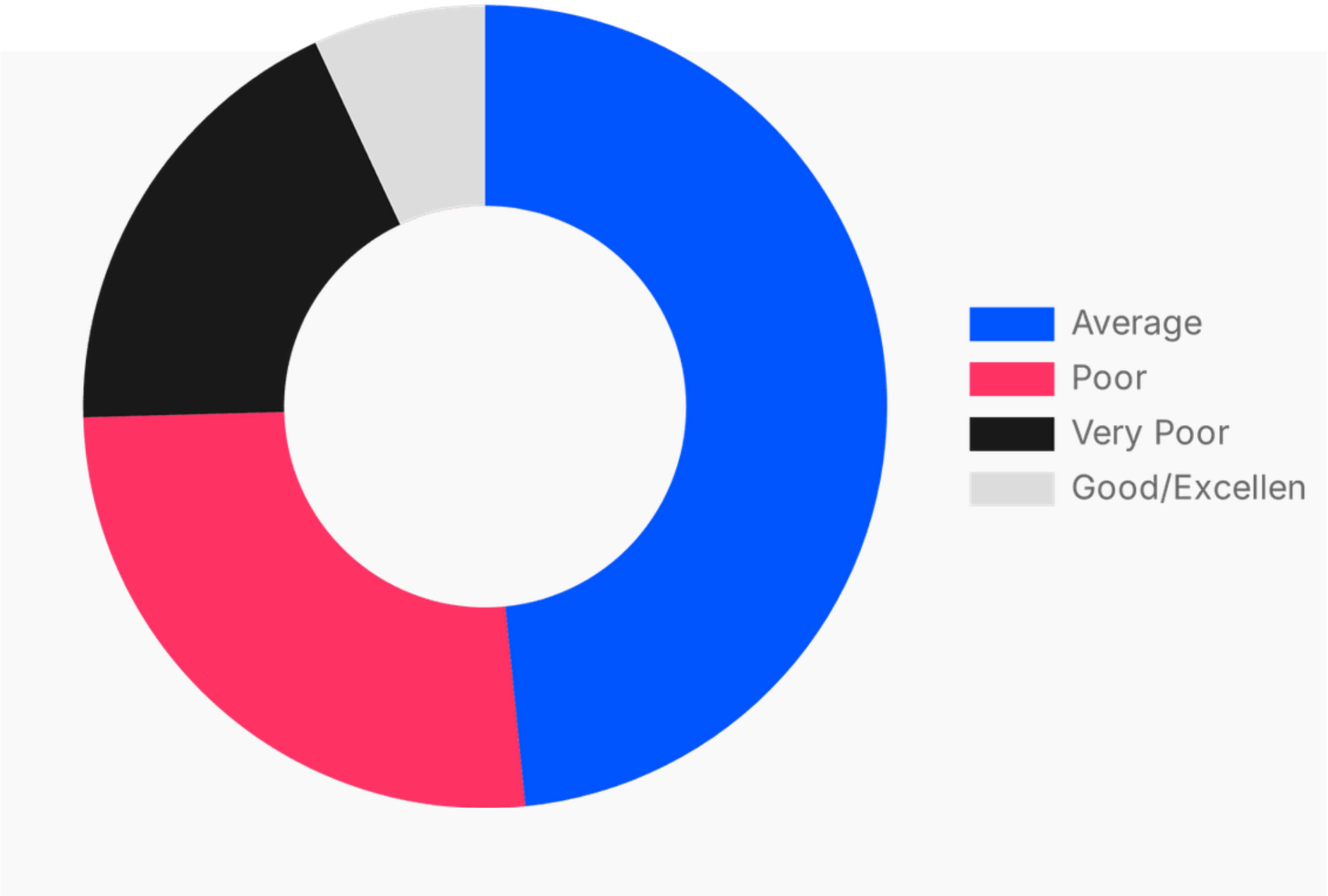
USER EXPERIENCE (QOE)



Despite heavy reliance on mobile networks...

HIGH USAGE BUT POOR EXPERIENCE

Overall Network Experience (%)



DOMINANT USER CONCERNS

82.8%

Of respondents identify SLOW INTERNET as their primary network challenge.



69%

Experience issues "Often" or "Very Often"

54%

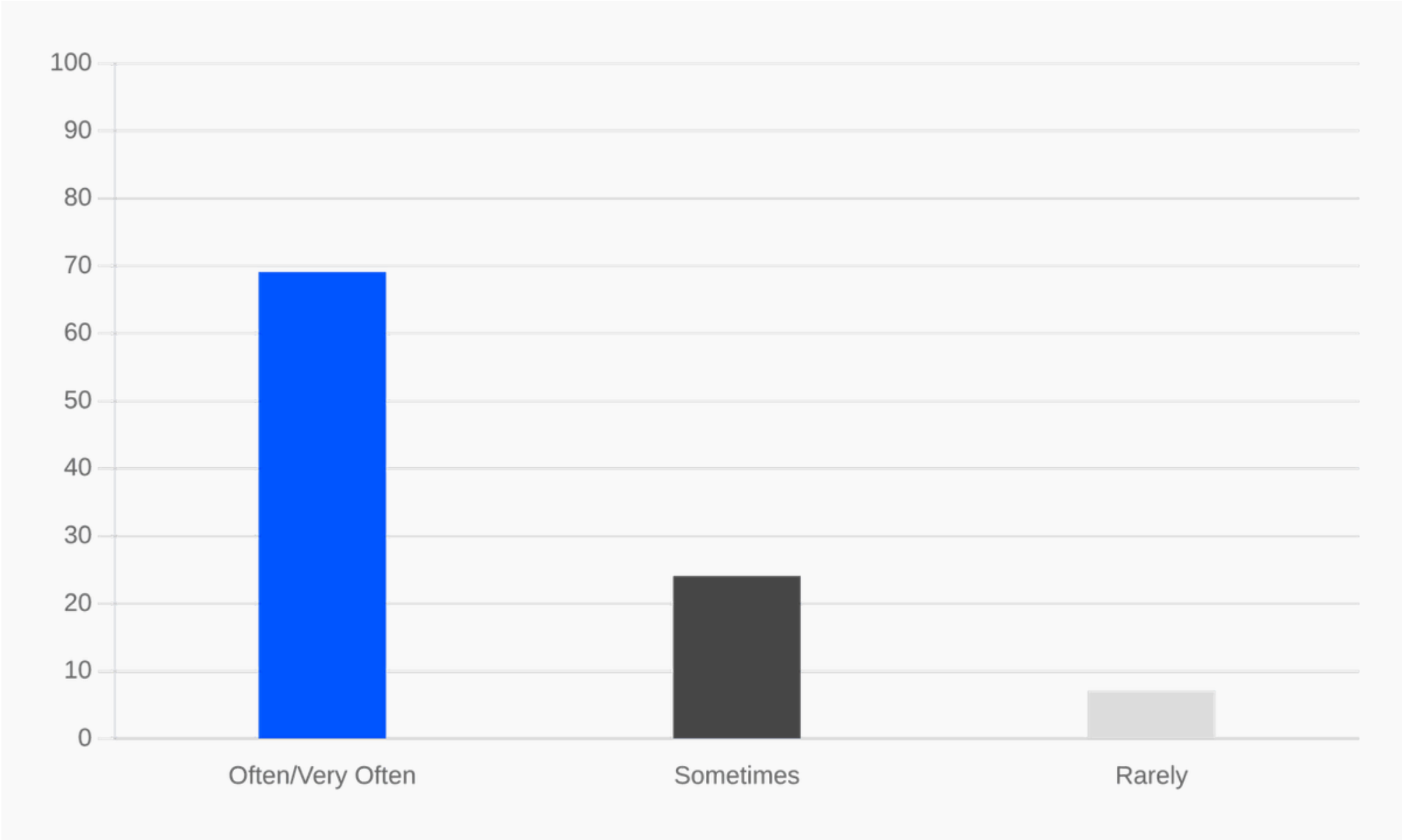
Report frequent video buffering (QoE Indicator)

60.7%

Encounter problems everywhere (Systemic Issue)

PROBLEM FREQUENCY

How often issues occur (%)

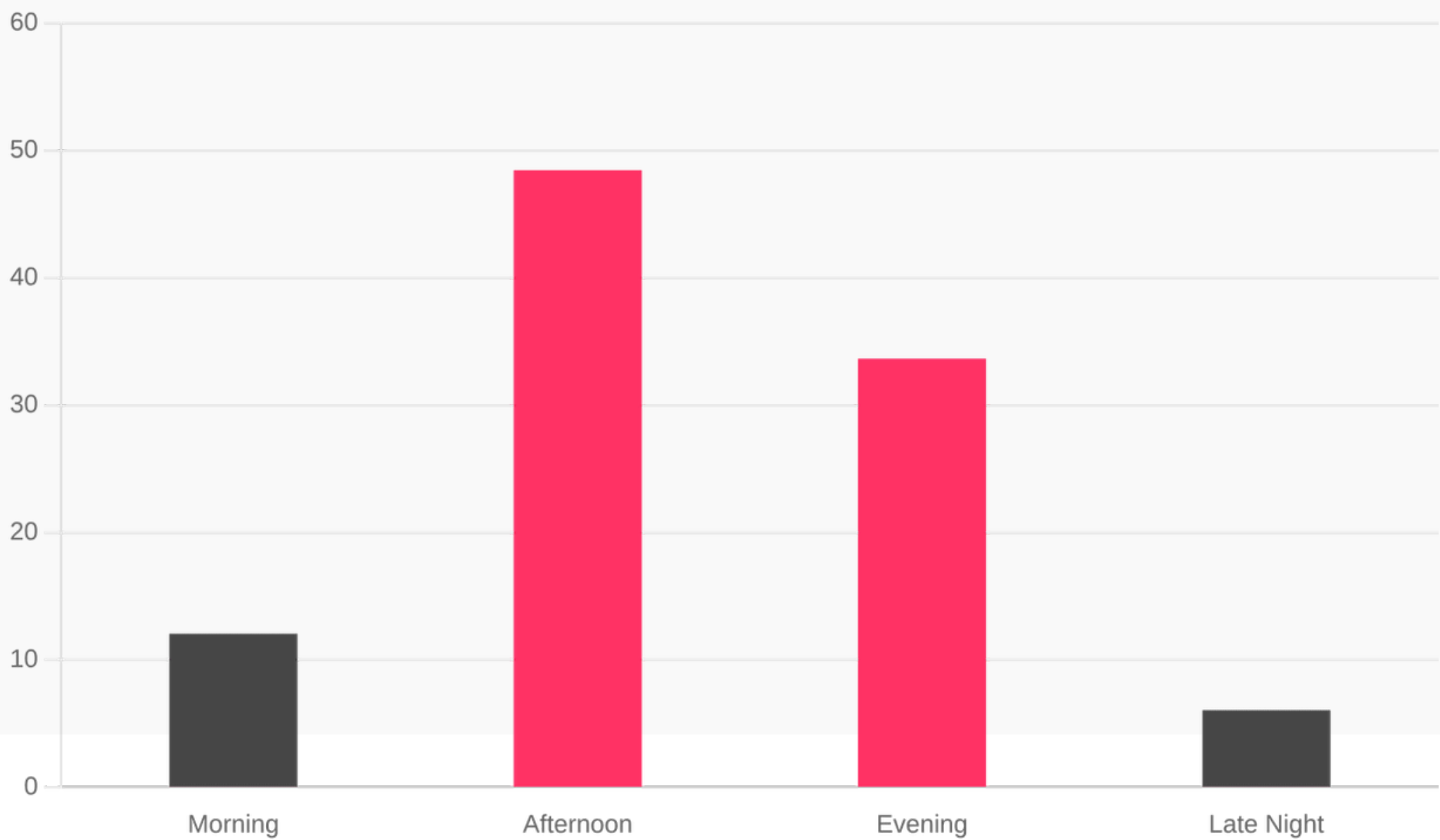


**PROBLEMS ARE
FREQUENT, NOT
OCCASIONAL**

69% face issues "Often"

CONTEXT INSIGHT

TIME OF WORST NETWORK PERFORMANCE (%)

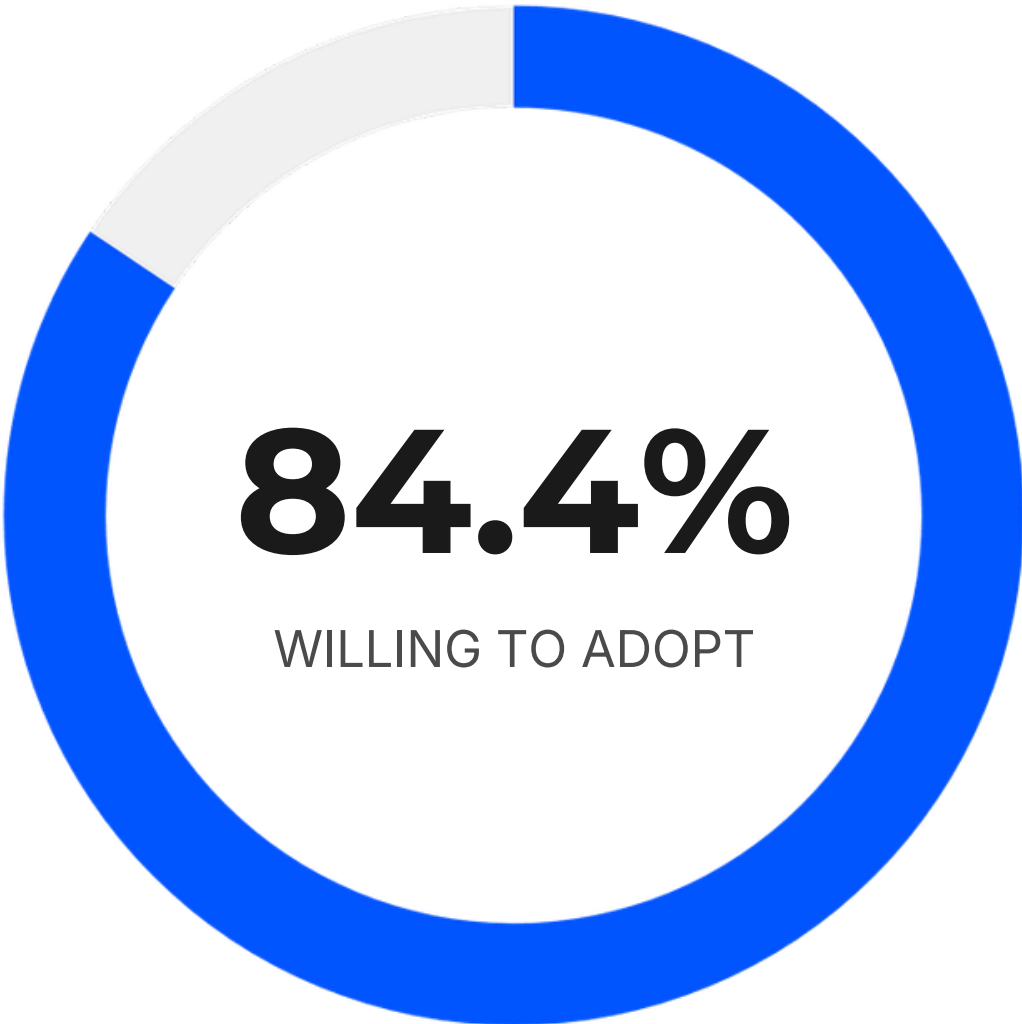


KEY OBSERVATION

“Likely network congestion periods”

Performance peaks in failure during afternoon and evening hours.

USER READINESS & ADOPTION



Survey Result: Interest in Network Performance Analysis App

**DATA SHARING**

76% of users are open to automatic data collection if the value proposition is clear.

**INTERACTION**

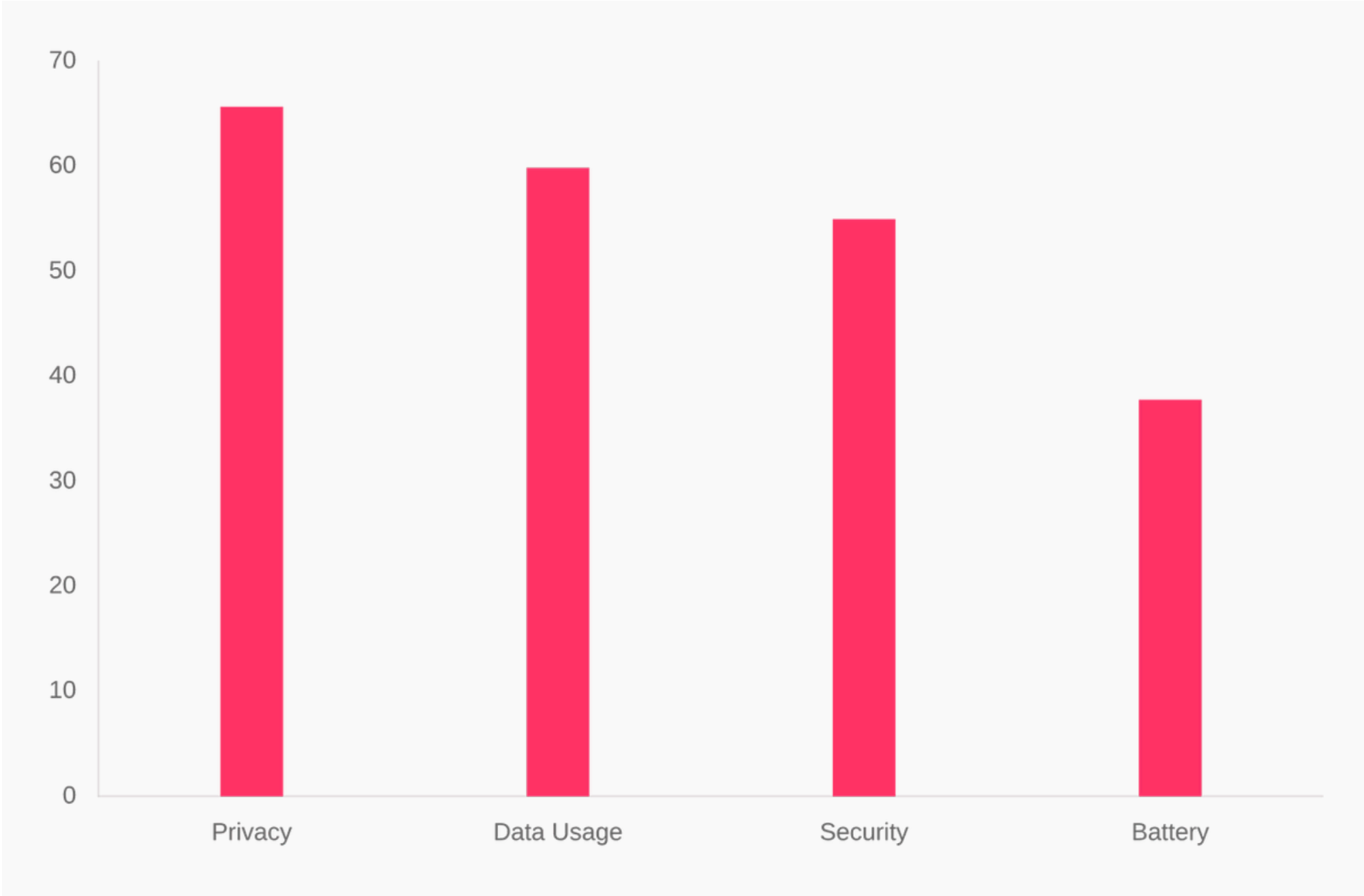
46.8% prefer event-based feedback (only when problems occur) over fixed intervals.

**TAKEAWAY**

Users demand intelligent, non-intrusive, and context-aware system interactions.

Platform User Concerns

Primary Concerns (%)



TRUST IS A MAJOR BARRIER

Privacy is the top concern

INTERVIEW INSIGHTS (MTN)



Real-time network monitoring systems already in place.



Internal service management processes are highly mature.



Detailed operational data is strictly confidential.

PROVIDER GAP

“They have QoS, but limited user-level visibility”

THE KEY GAP

USERS SAY:

“Poor overall experience”

“Frequent performance issues”

PROVIDERS SAYS:

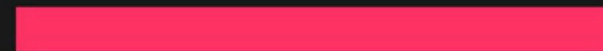
“Strong monitoring systems”

“Mature QoS management”

**THERE IS A GAP BETWEEN NETWORK PERFORMANCE
(QOS) AND USER EXPERIENCE (QOE)**

THEY HAVE THE NETWORK VIEW

**WE PROVIDE THE HUMAN
EXPERIENCE LAYER**



KEY INSIGHTS SUMMARY



High dependence on mobile internet



Poor overall user experience



Slow internet is the dominant issue



Network problems are frequent



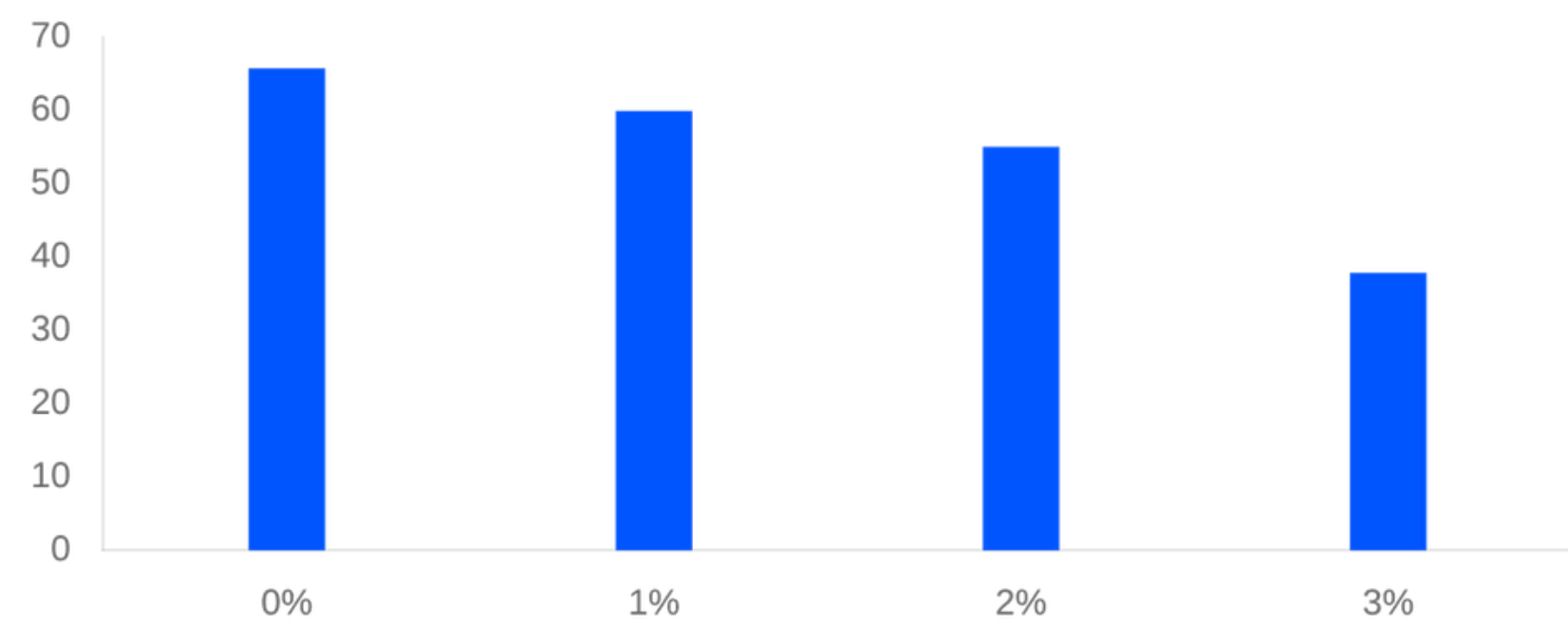
Users are willing to adopt the solution



Privacy concerns must be addressed

DESIGN CONSTRAINTS & RELUCTANCE

USER CONCERNS



Primary barriers to app adoption among surveyed users.

PROVIDER CONSTRAINTS



Data Confidentiality

Strict internal policies regarding network performance and user behavior data sharing.



Security & Compliance

Regulatory requirements and operational sensitivity prevent open data access.

System Mandate

The solution must be **Privacy-Conscious**, **Resource-Efficient**, and **Anonymized**.

CONCLUSION

- ✓ Data collected successfully
- ✓ Problem clearly validated
- ✓ Stakeholder needs identified

UP NEXT

REQUIREMENT ANALYSIS (SRS & SYSTEM DESIGN)